

Unit 1.2 The Absolute Value.

Source: Stewart, Appendix A pp A6 - A9.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the absolute value.
2. be able to write down and use the properties of the absolute value.
3. be able to solve inequalities with absolute values.
4. be able to sketch the graph of $y = |f(x)|$.

The absolute value of 'a' denoted by $|a|$ is the distance between 'a' and 0 on the number line, and $|a| \geq 0$ for all $a \in \mathbb{R}$ (a element of real numbers).

Definition

The absolute value of 'a' is defined as follows

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

(use the definition
to sketch graphs)

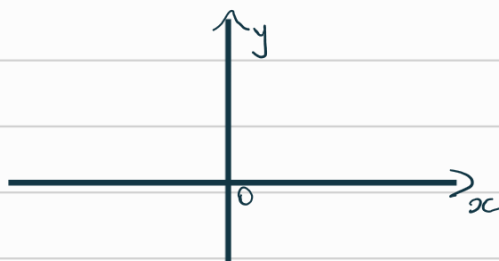
e.g

$$|-5| = \quad ; \quad |\pi - 3| =$$

$$|6| = \quad ; \quad |3 - \pi| =$$

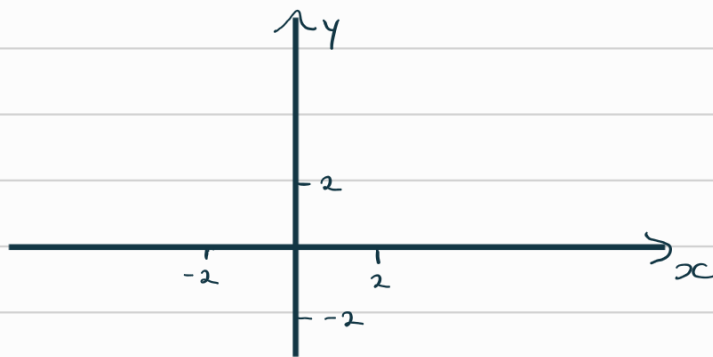
$$|2x - 3| =$$

Sketch the graph of $y = |x|$



Examples

- 1) Sketch the graph of the following
 $f(x) = |x+2|$



- 2) $f(x) = |x^2 - 1|$



- 3) $f(x) = |x| + x$



Properties of Absolute Value

$$\begin{array}{ll} \textcircled{1} & |ab| = |a||b| \quad ; \quad \textcircled{3} \quad |a^n| = |a|^n \\ \textcircled{2} & \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad \textcircled{4} \quad \sqrt{a^2} = |a| \end{array}$$

5 $|a| \geq 0$ for all $a \in \mathbb{R}$

6. $|a| = 0$ if and only if $a =$

P.A7(6) (use this to solve equations/inequalities)

Suppose $a > 0$. Then

1. $|x| = a$ if and only if
2. $|x| < a \iff$
3. $|x| > a \iff$

Examples

4 Solve the following (Solving $\Rightarrow$ finding a solution set)

a) $|x| < 3$

b) $|x| = -4$,

c) $|x^2 - 14| = 1$

d) $|2x - 3| \leq 0.4$

$$e) \left| \frac{2x-1}{x+1} \right| = 3$$

$$f) \frac{x}{|x-2|} > 0$$

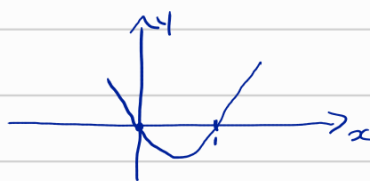
$$\frac{|x|}{|x-2|} > 0$$

$$g) 1 \leq |x| \leq 4$$

$$h) \frac{|x-6|}{x^2-x} \leq 0$$

$$\frac{|x-6|}{x^2-x} = 0 \quad \text{OR} \quad \frac{|x-6|}{x^2-x} < 0$$

$$x = 6 \quad \text{OR} \quad x^2 - x < 0$$



$$\therefore x \in (0, 1)$$

$$\Rightarrow x \in (0, 1) \cup \{6\}.$$

$$h) |x+3| = |2x+1|$$

$$i) |x-1| > -4$$

$$j) |x-1| < -4$$

$$k) |2-x| > 0$$